Objemová práca

$$\mathrm{d}w = -p_{ex}\,\mathrm{d}V$$

Dej				
izotermický (dT = 0)		**izobarický (dp = 0)**	**izochorický (dV = 0)**	**adiabatický (q = 0)**
vratný (rev)	*nevratný (irev)*			
$p_{ex} = p = \frac{nRT}{V}$ $\mathrm{d}w = -p\,\mathrm{d}V$ $\mathrm{d}w = -\frac{nRT}{V}\mathrm{d}V$ $w = -nRT\int_{V_i}^{V_f}\frac{\mathrm{d}V}{V}$ $w = -nRT\ln\frac{V_f}{V_i}$ $(w = nRT\ln\frac{p_f}{p_i})$	p_{ex} = konšt. $\mathrm{d}w = -p_{ex}\,\mathrm{d}V$ $\mathrm{d}w = -p_{ex}\int_{V_i}^{V_f}\mathrm{d}V$ $w = -p_{ex}\,(V_f - V_i)$ voľná expanzia: $p_{ex} = 0 \Rightarrow w = 0$	$p_{ex} = p$ = konšt. $\mathrm{d}w = -p\,\mathrm{d}V$ $\mathrm{d}w = -p\int_{V_i}^{V_f}\mathrm{d}V$ $w = -p\,(V_f - V_i)$ $w = -nR\,(T_f - T_i)$ w aj q majú vlastnosti stavovej funkcie	$\mathrm{d}w = 0$ $w = 0$ $\Delta U = q$ q má vlastnosti stavovej funkcie	$\mathrm{d}w = \mathrm{d}U$ $w = \Delta U$ ak C_V = konšt.: $w = C_V\,(T_f - T_i)$ $w = C_V\left(\frac{p_fV_f}{nR} - \frac{p_iV_i}{nR}\right)$ $w = \frac{C_V}{nR}(p_fV_f - p_iV_i)$ $w = \frac{C_V}{C_p - C_V}(p_fV_f - p_iV_i)$ $w = \frac{1}{\frac{C_p - C_V}{C_V}}(p_fV_f - p_iV_i)$ $w = \frac{p_fV_f - p_iV_i}{\gamma - 1}$ $w = \frac{nR(T_f - T_i)}{\gamma - 1}$ w má vlastnosti stavovej funkcie